

Sri Lanka Institute of Information Technology



REQUIREMENT SPECIFICATION DOCUMENT

Group MLB-B6G2 - 07

Web-Based Pet Care Management System

Name	IT Number
SAVINDU S D N	IT25100618
DE SILVA L P B	IT25101709
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BALASOORIYA B A H N	IT25103607
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1. System Purpose

The Web-Based Pet Care Management System is an online platform that helps pet owners, veterinary service providers, and pet care centre staff manage pet information, healthcare activities, and boarding operations efficiently. It centralises pet profiles, appointment and boarding bookings, medical records, vaccination/medication tracking, and business reporting, reducing reliance on paper records and manual scheduling.

Main objectives:

- Provide centralised storage for pet information and healthcare records.
- Allow pet owners to register and manage one or more pets.
- Allow pet owners to search for veterinarians and veterinary clinics.
- Provide an efficient online appointment booking system.
- Maintain complete medical history for registered pets.
- Manage prescriptions, medications, treatments, and vaccinations.
- Manage pet boarding operations, including facility allocation and daily care tracking.
- Provide reminders for appointments, vaccinations, and medications.
- Allow veterinary clinics to manage veterinarians, services, schedules, and appointments.
- Provide secure, role-based access to sensitive information.
- Improve communication between pet owners and veterinary service providers.
- Provide business analytics and revenue reporting to support decision-making.

2. Users

The system supports nine categories of users, each with specific responsibilities and access privileges.

2.1 Pet Owner

Manages their profile and detailed pet information (photo, medical history, allergies, special care instructions), veterinary books, grooming, and boarding appointments, views real-time boarding status, tracks medical records and vaccinations, downloads invoices and prescriptions, receives notifications, and provides feedback.

2.2 Veterinary Officer

Manages profile and availability, views appointments and authorised pet information, records consultations, diagnoses, treatments and vaccinations, issues digital prescriptions, and sets vaccination reminders. Medical record editing is restricted to authorised veterinary staff.

2.3 System Administrator

Manages user accounts and access control, monitors system activity, manages system settings, and generates system reports.

2.4 Inventory Manager

Adds and updates products, monitors stock levels, manages suppliers and purchases, tracks expired/low-stock items, and maintains inventory records.

2.5 Pet Care Center Manager

Oversees daily operations across appointments, staff, boarding, and grooming. Monitors operational dashboards (appointments, vaccinations, boarding occupancy, revenue, staff workload, customer

feedback), receives alerts for missed appointments and overdue vaccinations/check-outs, and generates centre reports.

2.6 Grooming Specialist

Manages availability and service details, views grooming appointments, updates appointment status, and records completed services.

2.7 Customer Service Executive

Manages communication with pet owners: handles appointment bookings across veterinary, grooming, and boarding services, records customer inquiries and complaints, sends automated notifications/reminders, and tracks the status of customer requests.

2.8 Pet Boarding Supervisor

Manages boarding operations: records check-in/check-out details, allocates cages/kennels, maintains daily care logs, monitors facility availability, and maintains a boarding history for each pet.

2.9 Business Owner

Oversees business performance: accesses analytics and revenue reports, monitors system security, and plans long-term growth features such as loyalty programs and online payments.

3. Functional Requirements

3.1 Pet Owner

- Register and manage their account and detailed pet profiles (photo, medical history, allergies, special care instructions).
- Book, reschedule, and cancel veterinary, grooming, and boarding appointments.
- View real-time boarding status and pet-care service information.
- View medical records, prescriptions, and vaccination records.
- Download invoices, receipts, medical bills, and prescriptions as PDF.
- Make payments and view payment history.
- Receive notifications and reminders; submit ratings and feedback.

3.2 Veterinary Officer

- Manage professional profile and availability.
- View scheduled appointments and authorised pet information.
- Record consultations, symptoms, diagnoses, and treatments.
- Issue digital prescriptions (medicines, dosage, vet's name) viewable/downloadable by owners.
- Record vaccinations and set automatic due-date reminders.
- View prior medical records; editing restricted to authorised veterinary staff.

3.3 System Administrator

- Create, update, activate, and deactivate user accounts.
- Manage user roles and permissions.
- Monitor system activities and security.

- Manage system information and handle complaints/inappropriate content.
- Generate system reports.

3.4 Inventory Manager

- Add, update, and remove inventory items and categories.
- Monitor stock availability and update stock levels.
- Identify low-stock items and manage inventory records.
- Generate inventory reports.

3.5 Pet Care Center Manager

- Manage centre information and available pet-care services.
- Manage staff and service schedules.
- View operational dashboards (appointments, boarding occupancy, revenue, staff workload, customer feedback, service trends).
- Receive alerts for missed appointments, overdue vaccinations, and boarding check-outs.
- View and manage pet-care appointments and billing; generate centre reports.

3.6 Grooming Specialist

- Manage profile, services offered, and pricing.
- Manage schedules and availability.
- View appointments and update appointment status.
- Record grooming service details/history and view relevant pet information.

3.7 Customer Service Executive

- Manage appointment bookings across veterinary, grooming, and boarding services, avoiding double bookings.
- Record and track customer inquiries, service requests, complaints, and special instructions.
- Send automated reminders (appointments, vaccination due dates, boarding check-in/out, payment confirmations) via email/SMS.
- Automatically notify pet owners of cancellations or rescheduled appointments.
- View and manage the status of all customer requests.

3.8 Pet Boarding Supervisor

- Record check-in details (arrival/departure dates, boarding type, special instructions, feeding schedule, medical conditions).
- Assign and manage cage/kennel/room allocation; track facility status (available, occupied, cleaning, maintenance).
- Maintain daily care logs (feeding, medication, exercise, behaviour) and alert veterinary staff when needed.
- Record check-out details (departure time, charges, comments) and maintain boarding history per pet.

3.9 Business Owner

- View business analytics dashboards (revenue by period/service, active/new customers, retention, demand trends).
- View revenue reports (payments received, outstanding payments, cancellations, refunds, profitability).

- Monitor system security settings (access control, password policies, activity logs).
- Review customer feedback analysis to support long-term growth planning.

4. Non-Functional Requirements

Performance

- Acceptable response times under normal load; fast search and appointment-availability lookups.
- Support multiple concurrent users with optimised database queries.

Security

- Authentication required for all private information; passwords stored securely (hashed).
- Role-based access control; medical records restricted to authorised users.
- Financial and revenue data restricted to authorised roles (e.g., Business Owner, Administrator).
- Secure session management and protection against unauthorised modification of records.

Reliability

- High availability with minimal downtime; consistent data storage with regular backups.
- Ability to recover from failures within an acceptable period.

Usability

- Simple, consistent, user-friendly interface; minimal effort to book appointments.
- Notifications clearly displayed; usable by people with basic computer knowledge.

Scalability

- Support a growing number of users, pets, clinics, records, and appointments without performance loss.

Maintainability

- Well-structured, organised database; components designed for easy maintenance.
- Errors and activities logged for troubleshooting; changes isolated from unrelated functions.

Availability

- Accessible via modern web browsers with minimal unexpected downtime.
- Planned maintenance communicated in advance; critical services available during operating hours.

Compatibility

- Support for common browsers and operating systems across desktop, tablet, and smartphone.
- Responsive interface across different screen sizes.

5. Constraints and Assumptions

5.1 Technical Constraints

- Requires a stable internet connection and modern web browser.
- Designed as a web-based system; performance depends on server resources and bandwidth.
- Requires a reliable database server with sufficient storage for medical/pet data.

5.2 Security Constraints

- Only authorised users may access confidential pet medical information.
- Valid credentials required for restricted functions; veterinarians must be verified before receiving privileges.
- Access to medical records controlled according to user role.

5.3 Project Scope Constraints

- Initial version focuses on pet-care, veterinary, and boarding management only.
- AI-based disease diagnosis/prediction is outside the initial scope.
- Integration with external veterinary labs, insurance providers, and emergency ambulance services is out of scope.
- Loyalty/membership programs, marketing automation tools, and product-sales integration are planned for future versions, not the initial release.

5.4 Assumptions

- Pet owners and veterinarians provide accurate information.
- Veterinary clinics have internet access; users have basic web-application literacy.
- Each user has a unique account; a pet may have multiple appointments and medical records.
- A veterinarian can treat multiple pets; a clinic can have multiple veterinarians.

6. Evidence of Requirement Gathering (Bot Interaction)

This section documents the requirement-gathering interactions conducted with a project discussion bot, simulating each stakeholder role, to elicit the requirements described in Sections 1-5 above.

6.1 Session 1 — Pet Care Center Manager (Nadeesha Perera)

Hi Nadeesha, as the Centre Manager for PawLife, what core operational features and reporting dashboards do you need to oversee daily clinic activities and staff performance?

 Project Discussion Bot - Year2

Nadeesha Perera (Pet Care Centre Manager):

Hello! As the Centre Manager, I need features that help me monitor all daily operations of the pet care centre in one place.

I would like to see appointment management, pet registration, customer management, staff schedules, vaccination reminders, boarding bookings, grooming services, and payment tracking. Staff should be able to update pet records and service status easily.

For dashboards, I need a simple summary showing:

- Today's appointments
- Upcoming vaccinations
- Pets currently in boarding
- Daily and monthly revenue
- Number of grooming, boarding, and veterinary services
- Staff attendance and workload
- Customer feedback and complaints
- Popular services and business trends

I also want alerts for missed appointments, overdue vaccinations, and boarding check-out dates. This helps me make quick decisions and ensure smooth operations.

These reports should be easy to view by day, week, or month.

Caption: Nadeesha highlighted the need for a unified operations dashboard (appointments, vaccinations, boarding, revenue, staff, feedback) and automated alerts for missed appointments, overdue vaccinations, and boarding check-outs — reflected in Sections 2.5 and 3.5.

6.2 Session 2 — Veterinary Officer (Kasun Jayawardena)

Hi Kasun, as the Veterinary Officer, what specific workflow features do you need for managing pet health profiles, recording vaccinations, and issuing digital medical records or prescriptions?



Project Discussion Bot - Year2

Kasun Jayawardena (Veterinary Officer):

Hello. For my daily work, I need each pet to have a complete health profile. This should include the pet's basic details, owner information, medical history, allergies, previous treatments, vaccination records, and any ongoing health conditions.

When a vaccination is given, I should be able to record the vaccine name, date, next due date, and notes. The system should also remind staff and pet owners when the next vaccination is due.

For medical records, I need to enter examination details, diagnoses, treatments, laboratory results, and follow-up notes. These records should be stored safely and be easy to find during future visits.

I would also like to issue digital prescriptions. The prescription should show the pet's details, medicines, dosage instructions, and the veterinarian's name. Pet owners should be able to view or download these records online.

It is important that only authorized veterinary staff can edit medical records, while other staff can only view information that is relevant to their work. This helps maintain accuracy and confidentiality.

Caption: Kasun detailed the need for complete pet health profiles, vaccination reminders, structured medical records, and digital prescriptions with role-restricted editing — reflected in Sections 2.2 and 3.2.

6.3 Session 3 — Customer Service Executive (Dilani Fernando)

Hi Dilani, as the Customer Service Executive, what requirements do you have for handling appointment bookings, managing customer inquiries, and sending notifications or reminders to pet owners?



Project Discussion Bot - Year2

Dilani Fernando (Customer Service Executive):

Hello! My main work is communicating with pet owners and managing their service requests.

I need a simple appointment booking system where customers can book veterinary visits, grooming sessions, and boarding services. It should show available dates and help avoid double bookings.

For customer inquiries, I would like a place to record questions, service requests, complaints, and special instructions from pet owners. This helps staff follow up properly.

Notifications are very important. The system should send reminders for upcoming appointments, vaccination due dates, boarding check-in and check-out dates, and payment confirmations. Customers should receive notifications by email or SMS.

If an appointment is cancelled or rescheduled, pet owners should be informed automatically. I would also like to see the status of all customer requests so I can respond quickly and provide better customer service.

Caption: Dilani's interview surfaced the need for unified booking across services, inquiry/complaint tracking, and automated multi-channel notifications — added as a new role in Sections 2.7 and 3.7.

6.4 Session 4 — Pet Boarding Supervisor (Ruwan Samarasinghe)

Hi Ruwan, as the Pet Boarding Supervisor, how should the system track pet check-ins, check-outs, cage/facility allocation, and daily care logs during boarding stays?

 Project Discussion Bot - Year2

Ruwan Samarasinghe (Pet Boarding Supervisor):

Hello! For boarding operations, I need the system to record each pet's check-in and check-out details accurately.

When a pet checks in, staff should enter the arrival date, expected departure date, boarding type, special care instructions, feeding schedule, and any medical conditions. The system should then assign an available cage, kennel, or room and mark it as occupied.

During the stay, staff should maintain a daily care log. This log should record feeding times, medication given, exercise activities, grooming, behavior observations, and any health concerns. If a pet becomes sick or requires special attention, staff should be able to add notes and alert the veterinary team.

For facility management, I want to see which cages are available, occupied, under cleaning, or under maintenance. This helps us avoid allocation mistakes.

At check-out, the system should record the departure time, boarding charges, and any comments about the pet's stay. It would also be helpful to have a boarding history for each pet so we can review previous stays and special requirements during future visits.

Caption: Ruwan's interview revealed detailed boarding workflow needs (check-in/out, cage allocation, daily care logs, facility status) not previously captured — added as a new role in Sections 2.8 and 3.8.

6.5 Session 5 — Business Owner (Tharushi Silva)

Hi Tharushi, as the Business Owner of PawLife, what business analytics, revenue reports, system security controls, and long-term features are most critical for the growth of your centre?

 Project Discussion Bot - Year2

Tharushi Silva (Business Owner):

Hello. From a business perspective, I need reports that show how the centre is growing and where improvements are needed.

For analytics, I would like to see daily, monthly, and yearly revenue, service-wise income (veterinary, grooming, and boarding), number of active customers, new customer registrations, customer retention, and service demand trends. These reports will help me make business decisions and plan future investments.

For revenue reports, I need information about payments received, outstanding payments, booking cancellations, refunds, and overall profitability. I would also like to compare performance across different periods.

Security is very important. The system should have secure login, role-based access control, password protection, and activity logs. Different users should only access information related to their responsibilities. Customer and pet medical information must be protected.

For long-term growth, I would like features such as online payments, customer self-service portals, mobile access, loyalty or membership programs, online pet health history access, customer feedback analysis, and marketing support tools. In the future, it would also be useful to integrate with pet product sales and additional pet care services.

My main goal is to improve customer satisfaction, increase operational efficiency, and support the long-term growth of PawLife Pet Care Centre.

Caption: *Tharushi's interview identified business analytics, revenue reporting, and security requirements, plus long-term growth features scoped out as future work (Section 5.3) — added as a new role in Sections 2.9 and 3.9.*

6.6 Session 6 — Pet Owner Perspective

As a pet owner, what self-service features do I need on the online portal to manage my pets' profiles, track ongoing boarding status in real-time, and download medical invoices?

 Project Discussion Bot - Year2

Pet Owner Perspective:

Hello! As a pet owner, I would like the online portal to be simple and easy to use.

I want to create and manage profiles for all my pets. Each profile should include my pet's name, breed, age, photo, medical history, vaccination records, allergies, and special care instructions. I should also be able to update this information when needed.

For boarding services, I would like to see my pet's status in real time. The portal should show whether my pet has checked in, its assigned boarding area, feeding updates, medication updates, exercise activities, and any notes added by staff. It would be helpful to receive notifications if there are important updates about my pet's health or wellbeing.

I also want access to my appointment history, vaccination schedules, and upcoming bookings. This will help me keep track of all pet care activities in one place.

For payments, I should be able to view and download invoices, receipts, medical bills, boarding charges, and prescription records in PDF format. I would also like to see my payment history and any outstanding balances.

Additionally, it would be useful if I could book appointments online, request boarding services, receive reminders, and communicate with the centre without having to call or visit in person. These features would make the service much more convenient for pet owners like me.

Caption: *This session detailed the self-service portal experience: pet profile management, real-time boarding status, downloadable invoices/prescriptions, and online booking/communication — reflected in Sections 2.1 and 3.1.*